ME315A_LAB03

AUTHOR
Fernanda Di Vincenzo 280867

Lab 03 - Dados relacionais e operações do tipo join

Pacotes

```
library(readr)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(leaflet)
```

1. Importação dos dados

1.1 Aeroportos

```
airports <- read_csv(
  "airports.csv",
  col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE)
)
```

```
Rows: 322 Columns: 5
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (3): IATA_CODE, CITY, STATE
dbl (2): LATITUDE, LONGITUDE

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
airports
```

```
# A tibble: 322 × 5
   IATA_CODE CITY          STATE LATITUDE LONGITUDE
```

```
   <chr>     <chr>          <chr>     <dbl>    <dbl>
 1 ABE       Allentown      PA         40.7    -75.4
 2 ABI       Abilene        TX         32.4    -99.7
 3 ABQ       Albuquerque    NM         35.0   -107. 
 4 ABR       Aberdeen       SD         45.4    -98.4
 5 ABY       Albany         GA         31.5    -84.2
 6 ACK       Nantucket      MA         41.3    -70.1
 7 ACT       Waco           TX         31.6    -97.2
 8 ACV       Arcata/Eureka  CA         41.0   -124. 
 9 ACY       Atlantic City  NJ         39.5    -74.6
10 ADK       Adak           AK         51.9   -177. 
# ℹ 312 more rows
```

1.2 Companhias aéreas

```r
airlines <- read_csv("airlines.csv")
```

```
Rows: 14 Columns: 2
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (2): IATA_CODE, AIRLINE

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
airlines
```

```
# A tibble: 14 × 2
   IATA_CODE AIRLINE
   <chr>     <chr>
 1 UA        United Air Lines Inc.
 2 AA        American Airlines Inc.
 3 US        US Airways Inc.
 4 F9        Frontier Airlines Inc.
 5 B6        JetBlue Airways
 6 OO        Skywest Airlines Inc.
 7 AS        Alaska Airlines Inc.
 8 NK        Spirit Air Lines
 9 WN        Southwest Airlines Co.
10 DL        Delta Air Lines Inc.
11 EV        Atlantic Southeast Airlines
12 HA        Hawaiian Airlines Inc.
13 MQ        American Eagle Airlines Inc.
14 VX        Virgin America
```

1.3 Voos: leitura em blocos com `readr`

```r
tamanho_chunk <- 1000000

estatisticas <- list()

i <- 1
```

```r
processar_chunk <- function(chunk, pos) {

  chunk <- chunk %>%
    select(DESTINATION_AIRPORT, ARRIVAL_DELAY)

  chunk <- chunk %>%
    filter(!grepl("^1", as.character(DESTINATION_AIRPORT)))

  chunk <- chunk %>%
    filter(
      !is.na(DESTINATION_AIRPORT),
      !is.na(ARRIVAL_DELAY)
    )

  resultado <- chunk %>%
    group_by(DESTINATION_AIRPORT) %>%
    summarise(
      soma_atrasos = sum(ARRIVAL_DELAY),
      numero_voos = n(),
      .groups = "drop"
    )

  estatisticas[[i]] <<- resultado

  i <<- i + 1
}
read_csv_chunked(
  "flights.csv",
  callback = SideEffectChunkCallback$new(processar_chunk),
  chunk_size = tamanho_chunk,
  progress = FALSE
)
```

```
── Column specification ────────────────────────────────────────────────────────
cols(
  .default = col_double(),
  AIRLINE = col_character(),
  TAIL_NUMBER = col_character(),
  ORIGIN_AIRPORT = col_character(),
  DESTINATION_AIRPORT = col_character(),
  SCHEDULED_DEPARTURE = col_character(),
  DEPARTURE_TIME = col_character(),
  WHEELS_OFF = col_character(),
  WHEELS_ON = col_character(),
  SCHEDULED_ARRIVAL = col_character(),
  ARRIVAL_TIME = col_character(),
  CANCELLATION_REASON = col_character()
)
ℹ Use `spec()` for the full column specifications.

NULL
```

```r
flights <- bind_rows(estatisticas) %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    soma_atrasos = sum(soma_atrasos),
    numero_voos = sum(numero_voos),
    .groups = "drop"
  ) %>%
  mutate(
    atraso_medio = soma_atrasos / numero_voos
  )

flights
```

```
# A tibble: 322 × 4
   DESTINATION_AIRPORT soma_atrasos numero_voos atraso_medio
   <chr>                      <dbl>       <int>        <dbl>
 1 ABE                        12874        2220         5.80
 2 ABI                         9077        2224         4.08
 3 ABQ                       106415       18953         5.61
 4 ABR                        -2227         657        -3.39
 5 ABY                         7501         864         8.68
 6 ACK                         2393         487         4.91
 7 ACT                        16392        1541        10.6
 8 ACV                        12259        1267         9.68
 9 ACY                        41744        3530        11.8
10 ADK                         -284          89        -3.19
# ℹ 312 more rows
```

2. Join entre `flights` e `airports`

2.1 Aeroportos em `flights` ausentes em `airports`

```r
anti_join(flights, airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
```

```
# A tibble: 0 × 4
# ℹ 4 variables: DESTINATION_AIRPORT <chr>, soma_atrasos <dbl>,
#   numero_voos <int>, atraso_medio <dbl>
```

2.2 Combinando as tabelas

```r
flights <- flights |>
  left_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

flights
```

```
# A tibble: 322 × 8
   DESTINATION_AIRPORT soma_atrasos numero_voos atraso_medio CITY          STATE
   <chr>                      <dbl>       <int>        <dbl> <chr>         <chr>
 1 ABE                        12874        2220         5.80 Allentown     PA
 2 ABI                         9077        2224         4.08 Abilene       TX
```

```
 3 ABQ                    106415        18953          5.61 Albuquerque    NM
 4 ABR                     -2227          657         -3.39 Aberdeen       SD
 5 ABY                      7501          864          8.68 Albany         GA
 6 ACK                      2393          487          4.91 Nantucket      MA
 7 ACT                     16392         1541         10.6  Waco           TX
 8 ACV                     12259         1267          9.68 Arcata/Eureka CA
 9 ACY                     41744         3530         11.8  Atlantic City NJ
10 ADK                      -284           89         -3.19 Adak           AK
# ℹ 312 more rows
# ℹ 2 more variables: LATITUDE <dbl>, LONGITUDE <dbl>
```

3. Quantidade de aeroportos por estado

```
airports |>
  count(STATE, name = "n_aeroportos") |>
  arrange(desc(n_aeroportos))
```

```
# A tibble: 54 × 2
   STATE n_aeroportos
   <chr>        <int>
 1 TX              24
 2 CA              22
 3 AK              19
 4 FL              17
 5 MI              15
 6 NY              14
 7 CO              10
 8 MN               8
 9 MT               8
10 NC               8
# ℹ 44 more rows
```

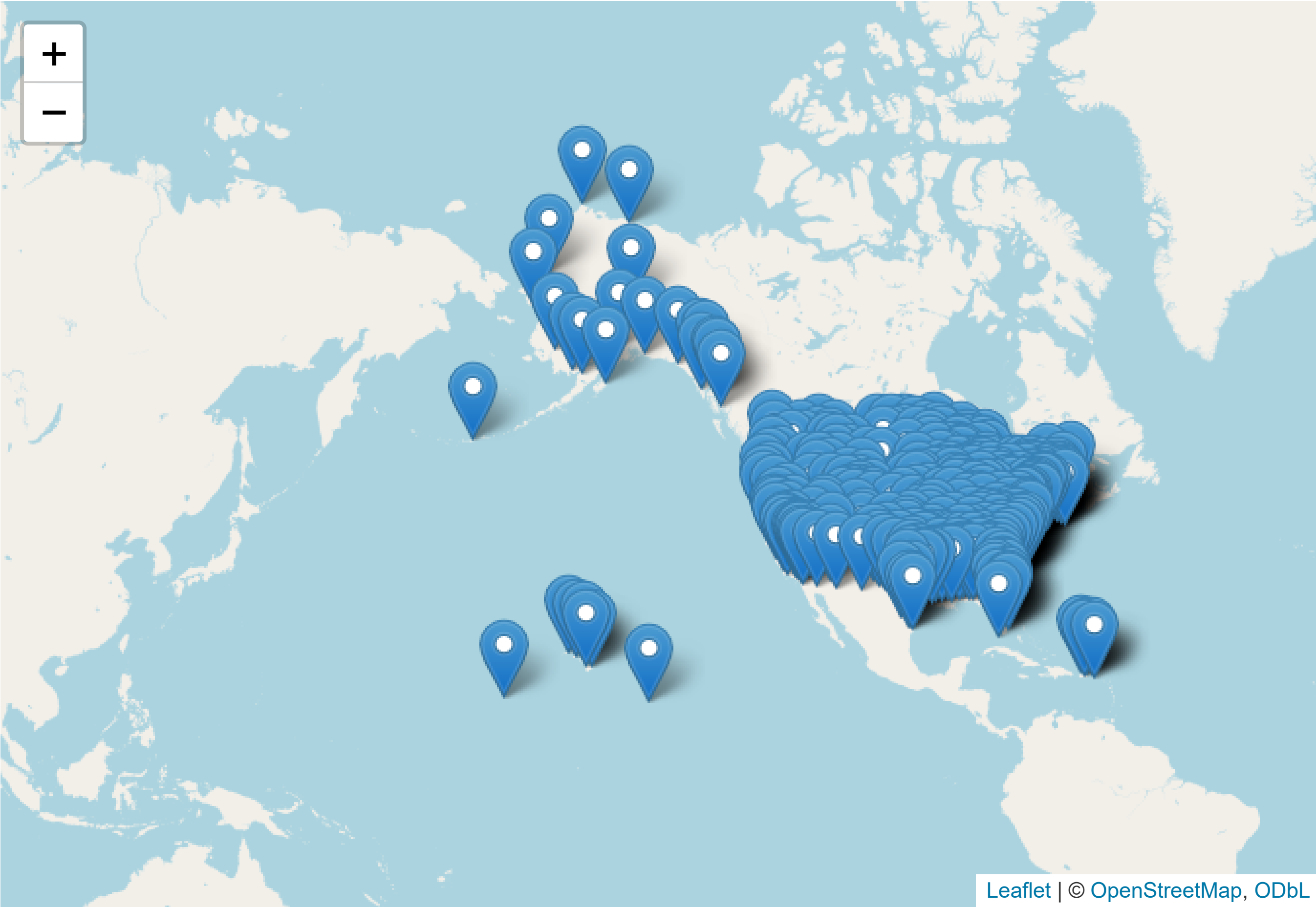
4. Mapa dos atrasos por aeroporto

```
dados_mapa <- flights %>%
  filter(
    !is.na(LATITUDE),
    !is.na(LONGITUDE)
  )

this <- leaflet(dados_mapa) %>%
  addTiles() %>%
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0(
      "<b>Aeroporto:</b> ",
      DESTINATION_AIRPORT,
      "<br><b>Cidade:</b> ",
      CITY,
```

```r
      "<br><b>Estado:</b> ",
      STATE,
      "<br><b>Atraso médio:</b> ",
      round(atraso_medio, 2),
      " minutos"
    )
  )

this
```




```r
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-31 21:38:22 -03"
```